
Appendix: High-Field Domains

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Abstract

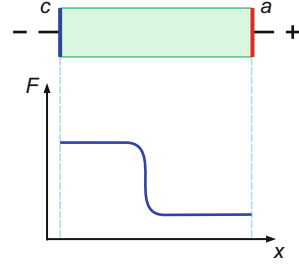
In regimes of over-linear negative differential conductivity of a bulk semiconductor, the field distribution must get inhomogeneous and divides into two parts: a domain with low conductivity and a high field, and the other part with a low field; this field distribution can be stationary or nonstationary. Both regions are joined by a narrow layer of only a few Debye width. The high-field domain can be imaged by applying the Franz-Keldysh effect. CdS has proved an ideal model semiconductor, where high-field domains can sweep the bulk free of space charges and connect the work functions of cathode and anode. The electrode-adjacent high field can set CdS into different thermodynamic stable states being *n*- or *p*-type, depending on whether the domain is attached to the cathode or anode. Within the domains, the crystal is free of space charges and permits to investigate sharp electronic transitions to defect levels around the quasi-Fermi levels without the interference from broadening electric fields.

Inhomogeneous Field or Current Distributions

Homogeneously doped semiconductors develop inhomogeneous distributions of carrier densities when this density at the metal-semiconductor boundary deviates from the density in the semiconductor bulk. Then, a space-charge layer is formed which creates inhomogeneous field and potential distributions near the electrode. When the deviation of the boundary electron density n_c from the bulk density n_{10} is significant, it influences the current through the semiconductor, reducing it by creating a Schottky barrier when $n_c \ll n_{10}$, and increasing it via carrier injection when $n_c \gg n_{10}$.

A different kind of inhomogeneous distribution of the electric field or the current density may occur in the regime of *nonlinear transport far from thermodynamic equilibrium*. In that case, the spatial patterns arise in a *self-organized* way, even in homogeneous semiconductors. Two types of nonlinear spatial patterns that are associated with specific field or current density inhomogeneities, respectively, are the topic of this appendix.

Fig. A.1 *Top:* Semiconductor between anode a and cathode c electrodes. *Bottom:* Electric field distribution in a semiconductor with a cathode-adjacent high-field domain



We will first ignore the boundary condition and deal with the semiconductor bulk. Close to thermodynamic equilibrium – that is, at sufficiently low bias voltage, the current density j as a function of the local electric field F in a homogeneous semiconductor is linear (*Ohm's law*), but at larger bias the $j(F)$ relation will generally become nonlinear and may display a regime of *negative differential conductivity*

$$\sigma_{\text{diff}} = \frac{dj}{dF} < 0. \quad (\text{A.1})$$

Thus, the current density decreases with increasing field and vice versa, which in general corresponds to an unstable situation. Here, the originally homogeneous field and current density distributions may evolve into an inhomogeneous pattern, referred to as a *high-field domain* sketched in Fig. A.1. The high-field domains can be stationary, when they are attached to an electrode, or they can move from one electrode to the other.

Negative Differential Conductivity

In bulk semiconductors, the drift-current density of electrons can generally be written as

$$j = e n(F) v(F). \quad (\text{A.2})$$

Here e is the electron charge, and $n(F)$ and $v(F)$ are the field-dependent electron concentration and drift velocity, respectively. The differential conductivity, assuming a homogeneous electric field, is expressed as

$$\sigma_{\text{diff}} = \frac{dj}{dF} = e \left(n \frac{\partial v}{\partial F} + v \frac{\partial n}{\partial F} \right). \quad (\text{A.3})$$

There are then two distinct ways to achieve $\sigma_{\text{diff}} < 0$ – that is, negative differential conductivity:

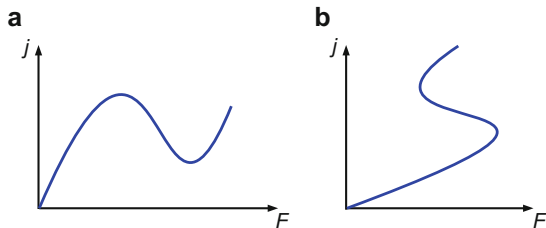


Fig. A.2 Typical current density-field characteristics that show a negative differential conductivity range. **(a)** N -shaped characteristics leading to high-field domain formation. **(b)** S -shaped characteristics leading to current filament formation

1. $dv/dF < 0$, which corresponds to negative differential mobility (*drift instability*)
2. $dn/dF < 0$, which corresponds to negative differential carrier concentration (*generation-recombination instability*)

We explore some typical example systems below.

Two important classes of negative differential conductivity (NDC) are described by an N -shaped or an S -shaped $j(F)$ characteristic and are denoted by $NNDC$ and $SNDC$, respectively (see Fig. A.2). However, more complicated forms like Z -shaped, loopshaped, or disconnected characteristics are also possible and may be treated analogously (Wacker and Schöll 1995). $NNDC$ and $SNDC$ are associated with voltage- or current-controlled instabilities, respectively. In the $NNDC$ case the current density is a single-valued function of the field, but the field is multivalued: the $F(j)$ relation has three branches in a certain range of j . The $SNDC$ case is complementary in the sense that F and j are interchanged. This duality is in fact far-reaching (Shaw et al. 1992).

The *global* current-voltage characteristic $I(V)$ of a semiconductor can, in principle, be calculated from the *local* $j(F)$ relation by integrating the current density j over the cross section A of the current flow

$$I = \int_A \mathbf{j} \cdot d\mathbf{a}, \quad (\text{A.4})$$

and the electric field F over the length L of the sample

$$V = \int_0^L F \, dx. \quad (\text{A.5})$$

Unlike the $j(F)$ relation, the $I(V)$ characteristic is not only a property of the semiconductor material, but also depends on the geometry, the boundary conditions, and the contacts of the sample. Only for the idealized case of spatially homogeneous states, the $j(F)$ and the $I(V)$ characteristics are identical, up to rescaling. The $I(V)$ relation is said to display *negative differential conductance* if

$$\frac{dI}{dV} < 0. \quad (\text{A.6})$$

In case of *NNDC*, the *NDC* branch is often but not always – depending upon external circuit, contacts, and boundary conditions – unstable against the formation of electric field domains, while in the *SNDC* case current filamentation generally occurs (Ridley 1963). These primary self-organized spatial patterns may themselves become unstable in secondary bifurcations leading to periodically or chaotically breathing, rocking, moving, or spiking filaments or domains, or even solid-state turbulence and spatiotemporal chaos (Schöll et al. 1998; Schöll 2001).

At this point, a word of warning is indicated. First, negative differential conductivity does not always imply instability of the steady state, and positive differential conductivity does not always imply stability. For example, *SNDC* states can be stabilized (and experimentally observed!) by a heavily loaded circuit, and, on the other hand, the bifurcation of a self-sustained oscillation can occur on a $j(F)$ characteristic with positive differential conductivity. Second, there is no one-to-one correspondence between *SNDC* and filaments or between *NNDC* and domains. Finally, it is important to distinguish between the local $j(F)$ characteristic and the global $I(V)$ relation, which may exhibit negative differential conductance even if the $j(F)$ relations do not show *NDC* and vice versa (Schöll 2001).

Domains and filaments can be theoretically described as special nonuniform solutions of the basic semiconductor transport-equations, subject to appropriate boundary conditions. A linear stability analysis around the spatially homogeneous steady state (singular point) for small space- and time-dependent fluctuations of the electromagnetic field and the relevant transport variables (e.g., carrier concentrations or charge densities) yields conditions for the onset of domain-type or filamentary instabilities (Schöll 1987).

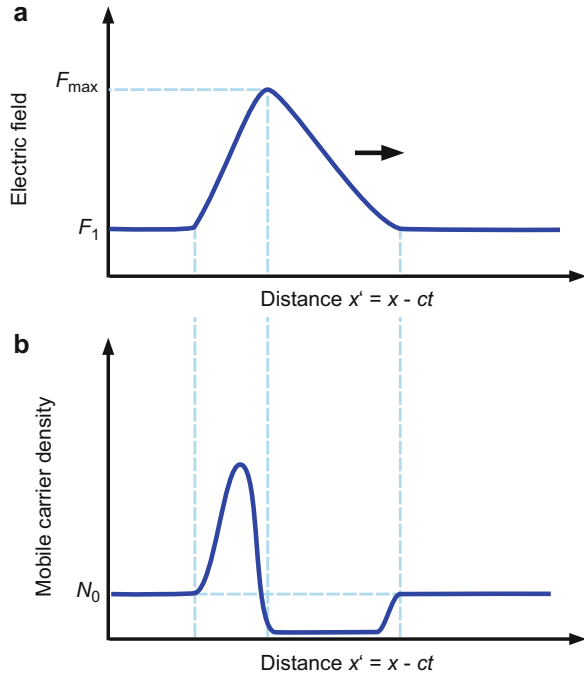
Domain Instability

The formation of field domains in a sample with *NNDC* can be understood from a simple argument. By way of example, we consider the case where the negative differential conductivity results from negative differential mobility, as in the Gunn effect (Gunn 1963, 1964), so that a range of fields exists where $dv/dF < 0$. The case of negative differential carrier concentration will be analyzed in Sect. [Phase-Space Analysis](#) below.

To illustrate the response of an *NNDC* element to a charge fluctuation, we consider a uniform field with a domain of increased field in the center of the sample as shown in Fig. [A.3a](#). The charge distribution that produces this field is shown in Fig. [A.3b](#). There is a net accumulation of charge on the left side of the domain and a depletion layer on the right. If we consider *negatively* charged carriers, the carriers and hence the domain will be moving to the right (cathode at left, anode at right).

Assuming that the field within the domain is within the *NDC* range and the field outside the domain is within the ohmic range, but close to the field of peak velocity, then it is clear that the field fluctuation will initially grow with time. This happens

Fig. A.3 (a) Electric field profile, (b) carrier density profile of a high-field Gunn domain. The domain is moving with velocity v in the positive x direction (schematic) (After Shaw et al. 1992)



because the higher upstream field in the center of the domain results in carriers moving more slowly than those at the edges, where the field is lower. Charge will therefore deplete on the right (leading) edge of the domain and accumulate at the left (trailing) edge. This charge will add to what is already there, increasing the field in the domain. If the element is in a resistive circuit, the increasing voltage across the domain will cause the current to decrease in the circuit and lower the field outside the domain. The field in the domain will continue to grow in the interior of the domain and drop outside. Thus, an instability of the uniform field with respect to domain formation results.

The constitutive equations in one spatial dimension are the carrier continuity-equation, neglecting recombination processes,

$$\frac{\partial n(x, t)}{\partial t} + \frac{\partial}{\partial x} \left(n(x, t) v(F) - D \frac{\partial n(x, t)}{\partial x} \right) = 0 \quad (\text{A.7})$$

and Gauss' law

$$\frac{\partial F(x, t)}{\partial x} = \frac{e}{\epsilon \epsilon_0} (n(x, t) - N_0), \quad (\text{A.8})$$

where eN_0 is the negative uniform background charge density, and D is the diffusion constant. Equations A.7 and A.8 can be combined by differentiating Eq. A.8 with

respect to time and substituting into Eq. A.7. Integration over x gives the total current density $j(t)$, which is composed of displacement current, drift, and diffusion contributions, respectively:

$$\varepsilon \varepsilon_0 \frac{\partial F(x, t)}{\partial t} + e \left(n(x, t) v(F) - D \frac{\partial n(x, t)}{\partial x} \right) = j(t), \quad (\text{A.9})$$

with the external current density $j(t)$. Substituting Eq. A.8 into Eq. A.9 to eliminate n yields the governing nonlinear transport equation

$$\varepsilon \varepsilon_0 \left(\frac{\partial F(x, t)}{\partial t} + v(F) \frac{\partial F(x, t)}{\partial x} - D \frac{\partial^2 F(x, t)}{\partial x^2} \right) + e N_0 v(F) = j(t). \quad (\text{A.10})$$

The stability of the uniform steady state F^* can be tested by linearizing Eq. A.10 around F^* for small space- and time-dependent fluctuations

$$\delta F(x, t) = \exp(\lambda t) \exp(ikx). \quad (\text{A.11})$$

Substituting this ansatz into Eq. A.10 for fixed external current density j determines the time increment λ for any given wave vector k – that is, the dispersion relation

$$\lambda = -\frac{1}{\tau_M} - D k^2 - i k v(F^*). \quad (\text{A.12})$$

Here the effective differential dielectric relaxation time $\tau_M \equiv (e N_0 (dv/dF)/\varepsilon \varepsilon_0)^{-1}$ has been introduced. For negative differential mobility ($dv/dF < 0$), τ_M is negative, and small fluctuations

$$\delta F(x, t) = \exp \left(-t \left(\frac{1}{\tau_M} + D k^2 \right) \right) \exp(ik(x - vt)) \quad (\text{A.13})$$

grow in time for $k^2 < (D |\tau_M|)^{-1}$ – that is, for long wavelengths. Equation A.13 describes an undamped traveling wave propagating in the positive x direction with velocity $v_0 = v(F^*)$.

This instability leads to the bifurcation of moving field domains from the uniform steady state, which depends, however, sensitively upon the sample length, the boundary conditions at the cathode contact, and the external circuit (Shaw et al. 1979, 1992). A simple criterion for the onset of a moving field domain can be derived from Eq. A.13 by neglecting diffusion and assuming that the linearization remains valid throughout its transit across the element of length L . The growth at the end of the transit – that is, after the transit time $t = L/v_0$ – is then given by

$$G = \frac{\delta F(t)}{\delta F(0)} = \exp \left(\frac{L}{v_0 |\tau_M|} \right) = \exp \left(\frac{N_0 L e \frac{dv}{dF}}{\varepsilon \varepsilon_0 v_0} \right). \quad (\text{A.14})$$

Substantial growth, i.e., a moving domain instability, occurs if the exponent in Eq. A.14 is larger than unity or

$$N_0 L > \frac{\varepsilon \varepsilon_0 v_0}{e \frac{dv}{dF}}. \quad (\text{A.15})$$

This is the $N_0 L$ product stability-criterion for the Gunn effect derived by McCumber and Chynoweth (1966) and by Kroemer (1968).

The fully developed field domains must be calculated from the full nonlinear transport equation (Eq. A.10), subject to appropriate boundary conditions. A simple visualization of the moving domain solution can be obtained by transforming Eq. A.10 to the comoving frame $x' = x - ct$, where c is the domain velocity, yielding

$$\varepsilon \varepsilon_0 \left((v(F) - c) \frac{\partial F}{\partial x'} - D \frac{\partial^2 F}{\partial x'^2} \right) + e N_0 (v(F) - v_\infty) = 0. \quad (\text{A.16})$$

Here $j = e N_0 v_\infty$ with $v_\infty = v(F_1)$ is the current density in the neutral material outside the domain. The result can be written in the form of a “dynamic system” of two first-order differential equations for F and $n = (\varepsilon \varepsilon_0 / e) \partial F / \partial x + N_0$, where the parameter corresponding to “time” is given by x' .

The topology of the solutions may be conveniently discussed in terms of a *phase-space analysis* or *field-of-directions analysis* (Shaw et al. 1992), which gives qualitative insight into the nature of the solutions and the effect of boundary conditions. Such methods have been extensively used by Böer and Voss (1968a) and Böer (1969) in the early investigations of Gunn domains and recombination domains, and they will be reviewed in the next section.

Phase-Space Analysis

The analysis of (stationary) high-field domains can be performed by inserting the field dependence of the relevant parameters (e.g., of the generation or recombination rates) into the set of governing transport equations; the solutions then can be obtained numerically.

Another type of analysis is suited to obtain more general information; it shows the solution type more easily and can be used to obtain quantitative information in a simple but satisfactory approximation. This analysis is known as *field-of-directions analysis* or as *phase-space analysis* and can be used easily in a *one-carrier model* in which the set of governing equations (transport, Poisson, and continuity equations) may be written as¹

¹For an n -type semiconductor, and assuming $p \ll n$ and $j_p \ll j_n \cong j$.

$$\frac{\partial n}{\partial x} = \frac{e}{kT} \left(\frac{j}{e\mu_n} - nF \right) \quad (\text{A.17})$$

and

$$\frac{\partial F}{\partial x} = \frac{e}{\varepsilon \varepsilon_0} (p_d - n - n_a). \quad (\text{A.18})$$

For definiteness we will assume first that the field dependence enters via reaction kinetics and makes the carrier density n , the density of electrons in acceptors n_a , and of holes in donors p_d implicit functions of F .

The solution curves of interest are $n(x)$ and $F(x)$ and, derived from the latter, the electron potential distribution $\Psi_n(x)$. Because Eqs. A.17 and A.18 are an autonomous system of differential equations that does not contain x explicitly, and fulfills the Lipschitz conditions of having continuous partial derivatives with respect to F and n , the solution curves can be projected into any nF plane, yielding a single-valued curve. That is, through every point of the nF plane traverses one and only one solution curve or every point in this plane uniquely defines *one* slope, except for *singular points* or *fixed points* which are defined by

$$\frac{dF}{dx} = \frac{dn}{dx} \equiv 0. \quad (\text{A.19})$$

Solutions cannot leave or enter singular points that represent solutions in which $n(x) = \text{const}$ and $F(x) = \text{const}$. One therefore concludes that no solutions can cross each other in the entire nF plane.

When identifying the direction of increasing x from the cathode toward the anode, one can attach to each point in the nF plane an arrow indicating this direction as illustrated in Fig. A.4. This field of arrows is known as a *field of directions*.

When the boundary values [$n(x=0)$ and $F(x=0)$] are given, one can start from this point and follow along the directions from point to point to obtain the solution curve $n(F)$. Because an x value is attached to every point in the nF plane, one finally obtains the desired $n(x)$ and $F(x)$. Such a procedure seems to be tedious. However, with the help of two auxiliary curves that can easily be identified, a simple method can be derived, which permits immediate classification of the possible set of solution curves. These auxiliary curves (*null-isoclines*) are those for which (Böer and Wilhelm 1963)

$$\frac{dF}{dx} \equiv 0, \text{ identified as } n_1(F), \quad (\text{A.20})$$

and

$$\frac{dn}{dx} \equiv 0, \text{ identified as } n_2(F), \quad (\text{A.21})$$

and are shown in Fig. A.5.

Fig. A.4 Field of directions computed for a typical homogeneous semiconductor, with the neutrality electron density $n_1(F)$ of vanishing space charge, and the drift-current curve $n_2(F)$ of vanishing diffusion current. $n_1(F)$ and $n_2(F)$ divide the nF plane into four quadrants I to IV, which indicate by the arrows the different signs of dF/dx and dn/dx

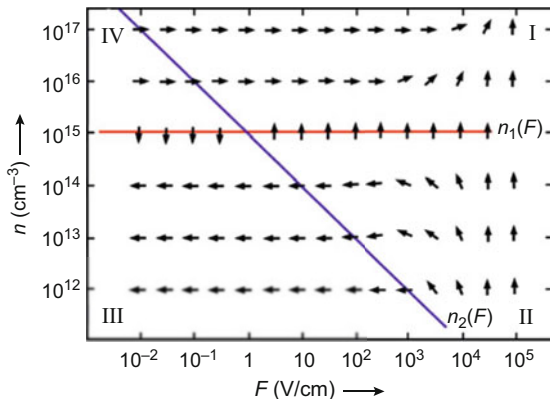
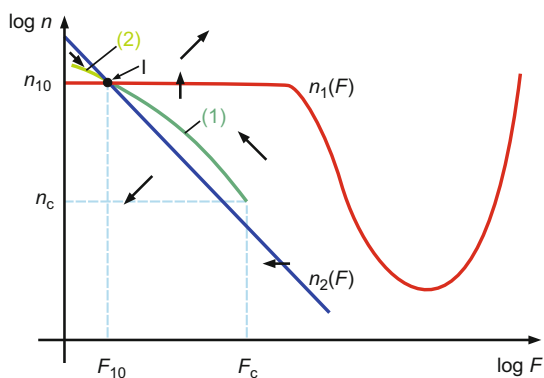


Fig. A.5 Symbolized field of directions with quasi-neutrality curve $n_1(F)$ and drift-current curve $n_2(F)$; solid arrows indicate the four quadrants of possible directions of the solution curve. I indicates the singular point; curves (1) and (2) show typical solutions for a Schottky barrier and an injection contact, respectively



The first curve, $n_1(F)$, is known as the *quasi-neutrality curve* because, following the Poisson equation, the space charge vanishes for any point on this curve. It can be crossed only vertically in the nF plane because, according to Eq. A.20, F remains constant with changing x . If F changes in x , the $n_1(F)$ curve is the locus of field extrema.

The second curve, $n_2(F)$, is known as the *drift-current curve* (see Eq. A.17), since for any point on this curve the diffusion current vanishes. It is therefore also referred to as *diffusion-neutrality curve*. This curve can only be crossed horizontally in the nF plane because n remains constant with changing x according to Eq. A.21.

Below $n_1(F)$, one obtains² $dF/dx < 0$; above $n_1(F)$, one has $dF/dx > 0$, as seen from Eq. A.18. In a similar fashion, one sees that $dn/dx < 0$ below and > 0 above $n_2(F)$. Therefore, the direction of any solution curve $n(F)$ points into different

²Observe that the field F is negative here.

quadrants below or above the quasi-neutrality and the drift-current curves, as symbolized by the arrows³ in Fig. A.5. This figure also shows typical shapes for the $n_1(F)$ and $n_2(F)$ curves.

The $n_1(F)$ curve shows the field dependence of the electron density under the neutrality condition, that is, in the given one-carrier model $n_1(F)$ is the same as $n(F)$ in a homogeneous and space-charge-free semiconductor. It is constant for low fields and usually increases at sufficiently high fields because of field excitation. With optical carrier generation it may have an intermediate field range in which n decreases with increasing F because of field quenching (Dussel and Böer 1970). Such a decreasing branch is included in the example given in Fig. A.5.

The $n_2(F)$ curve is a hyperbola for a given j , as long as the mobility is field independent. In the double logarithmic representation of Fig. A.5, it is given as a straight line with a slope of -1 . The crossing point of curves $n_1(F)$ and $n_2(F)$ is the singular point I. Solution curves that are long compared to the Debye length must approach this singular point. In the bulk of a semiconductor, which is usually several orders of magnitude thicker than the Debye length, $n(x)$ and $F(x)$ change very little with x ; hence, the solution curve “approaches” the singular point. It should, however, be remembered that no solution can leave or cross a singular point. Thus, a physically meaningful solution will always start or end in one of the quadrants, but slightly removed from the singular point.

One value of the field-of-direction analysis lies in the fact that one can immediately identify what types of solutions are possible. These are given as curves 1 and 2 and are the only types possible as stationary solutions for a long (compared to the Debye length) device.⁴

A typical solution curve for a Schottky barrier is shown as curve 1 in Fig. A.5. It starts at the metal-semiconductor interface ($x = 0$) at a lower density n_c and a higher field F_c than in the bulk (given by the singular point) and follows the field of directions (see Fig. A.4) toward the singular point I. Another solution is shown for an injecting contact as curve 2 in Fig. A.5. It starts at a higher electron density and consequently at a lower field at $x = 0$, and it also moves toward the singular point I.

From Fig. A.5 one sees that it is possible to approach the singular point I from the second or fourth quadrants only; it is impossible to enter any other quadrants from the first or third quadrant. Namely, crossing the $n_1(F)$ or $n_2(F)$ curve from the second or fourth quadrant means irrevocably turning away from the singular point, yielding physically meaningless solutions. In mathematical terms the singular point I is a saddle point, i.e., it is impossible to circle it.

³The arrow in the upper right quadrant, for example, indicates that through any point in this quadrant the solution curve must have a direction between 0° and 90° ; this holds correspondingly for the other quadrants.

⁴These solutions represent a device with one electrode. When including the other electrode one must be careful with the signs in the transport equations which result in a reversal of the arrows in the field of directions when approaching the second electrode; this reversal permits the traverse of the projection of the solution curve in the opposite direction in part of the solution.

We will now apply this field-of-direction analysis to include a range of negative differential conductivity. This can be achieved by a decrease of the carrier density due to field quenching or a decrease of the mobility, for example, by carrier excitation into a sidevalley in GaAs, with increasing field.

Decrease of Carrier Density with Field

The carrier density in a homogeneous semiconductor can decrease with increasing field only in nonequilibrium conditions. In steady state, this means that the field causes an increase of the recombination (a reduction of the generation is unlikely). This can be achieved in a number of ways; most researched are quenching transitions in which the field activates fast recombination centers.

Field Quenching

Field quenching is a means to reduce the space charge by field-enhanced ionization (Frenkel-Poole ionization) of minority carriers from compensating deep hole traps. When incorporating fast recombination centers (N_r), the model can explain field quenching of photogenerated carriers in homogeneous photoconductors.

Figure A.6 shows such a model. Light generates electron-hole pairs; when the holes are captured more readily by Coulomb-attractive hole traps (N_a) that have a very small capture cross section for electrons, the electron density in the conduction band increases substantially and n -type photoconductivity is observed: The lifetime of photogenerated electrons is much larger than that of holes, thus $n \gg p$. When a sufficient field is applied, however, the trapped holes can be freed from these Coulomb-attractive hole traps. A fraction of them will be captured by fast recombination centers (Fig. A.6b), which have a much larger cross section for electrons (typically 10^{-16} cm^2) than the deep hole traps (typically 10^{-21} cm^2). This causes a reduction of the electron density as a function of the field. Such a reduction of $n(F)$ is pictured in the quasi-neutrality curve $n_1(F)$ of Fig. A.5 and immediately precedes the steeply rising branch of carrier generation by field ionization.

Cathode- and Anode-Adjacent Domains

When increasing the bias to the device, the current increases, thereby causing a parallel shift of the drift current curve toward higher values – for example, from curve (1) to curve (2) in Fig. A.7. If the field-quenching branch of the quasi-neutrality curve decreases more than linearly with the field, then $n_1(F)$ and $n_2(F)$ will cross again, thereby creating two more singular points (II and III in Fig. A.7).

Assuming for simplicity that the boundary density n_j does not change with increasing bias and current ($n_j = n_c$), the field at the boundary increases until the starting point of the solution has moved close to the second singular point. Then the drift-current curve cannot rise further, because it would force the solution curve to cross $n_1(F)$ while maintaining the boundary condition at n_c . Such a crossing of $n_1(F)$ from the fourth quadrant is impossible. This means that with further increasing bias, $n_1(F)$ can no longer rise, the current must saturate, and the solution expands in x near

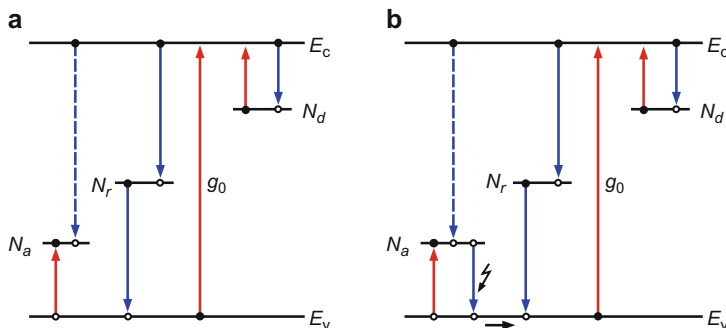
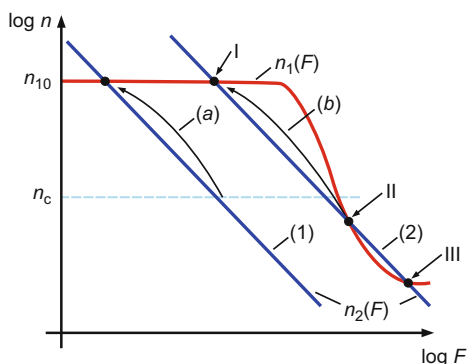


Fig. A.6 (a) Band model of an n -type photoconductor with hole traps N_a and fast recombination centers N_r . (b) Field-induced transition from slow (hole traps) to fast recombination centers causes field quenching of the photoconductivity

Fig. A.7 Field of directions representation with quasi-neutrality curve (n_1) and two drift-current curves (n_2) for lower (1) and higher (2) current with corresponding solution curves (a) and (b)



the second singular point. This solution (2) becomes *step like*, i.e., $n(x)$ and $F(x)$ are almost constant near $x = 0$, rather than $n(x)$ steeply increasing and $F(x)$ linearly decreasing near $x = 0$ in a Schottky barrier. This distinctly different, step-like behavior of $n(x)$ and $F(x)$ is referred to as a *high-field domain* (Böer et al. 1958; Böer 1959; Böer and Wilhelm 1963).

With increased bias, the high-field domain widens without increasing the field at the cathode (in n -type materials), while the current remains constant. In the field-of-directions representation, no changes are visible, except for a minute extension of the solution closer to the singular point II.

As long as the field remains constant within the domain, here also the space charge vanishes. When the field is increased to initiate field quenching, localized holes in deep hole traps, which produce this space charge in compensated photoconductors, are freed and can either move into the adjacent electrode or recombine,

thus reducing the positive space charge. With sufficient depletion of these hole traps, the space charge is reduced to zero.⁵

When the bias is increased sufficiently so that the high-field domain has expanded to fill the entire device, the solution remains at the singular point II. A further increase in bias causes a minute increase of the current [a slight shift upward of $n_2(F)$] that brings the solution into the quadrant between the singular points II and III and between $n_1(F)$ and $n_2(F)$; it extends the solution from II toward the singular point III. A still further increased bias causes part of the solution (toward the anode) to approach III, now forming an anode-adjacent domain of even higher fields (F_{III}), which then expands from the anode toward the cathode until it fills the entire device.

In the next section, we will show that a major part of this domain behavior can be experimentally confirmed before electric breakdown phenomena set in. Without a field-of-direction analysis, this domain behavior would be rather difficult to explain. The given discussion, however, even though qualitative makes such analysis transparent. It is for this reason that we have included the more qualitative field-of-directions method with the otherwise preferred numerical or analytical discussion.

We will also show in the next sections that in a number of realistic cases this method provides enough information to yield unambiguously the carrier density and mobility as a function of the electric field from rather simple experiments. The key to the success of this simple analysis is the fact that the field is essentially constant within a domain, and the transition region between bulk and domain has a thickness of only a few Debye lengths; that is, its width can usually be neglected compared to the thickness of the device and the domain. Then a simple step approximation is sufficient to describe the voltage drop across the device:

$$V = F_{II}x_1 + F_I(d_1 - x_1). \quad (\text{A.22})$$

With changing bias the width x_1 of the high-field domain changes linearly. This permits the determination of F_I and F_{II} when the domain width is measured as function of the bias.

Franz-Keldysh Effect to Directly Observe High-Field Domains

The domain width can be measured directly with probing light by using the Franz-Keldysh effect (Franz 1958; Keldysh 1958), which causes the absorption edge to shift toward longer wavelengths with increasing field. When observing the light transmitted through the semiconductor platelet,⁶ one detects the domain as region of increased absorption when using light at the absorption edge (Böer et al. 1958, 1959), as shown in Fig. A.8. Such high-field domains were first seen in CdS (Böer et al. 1958) and have since been extensively investigated with this method.

⁵Overcompensation is avoided because a reversal of space charge would reduce the field that was the cause for field quenching and would automatically cause an adjustment to yield $\rho \equiv 0$.

⁶For reasons of optical observation, the geometry of the device was chosen so that the thin platelet dimension is parallel to the probing light and the field direction is perpendicular to it.

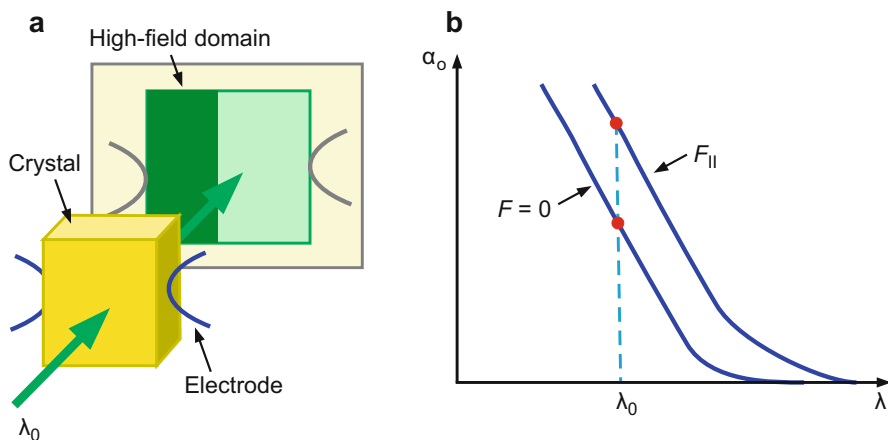


Fig. A.8 (a) Experimental setup to use the Franz-Keldysh effect for observing high-field domains. (b) Shift of the absorption edge due to the electric field. The crystal platelet is viewed in transmitted monochromatic light (λ_0) that shows the high-field domain as a region of increased absorption

The high-field domains are directly visible in the shadow projection given in Fig. A.9a for a CdS sample, showing the widening of the cathode-adjacent domain (upper row, at left) with increasing bias until the domain fills the entire device at 1750 V (upper row at right). With further increased bias, an anode-adjacent domain of higher field strength (with an increased band-edge shift, hence darker shadow) emerges from the anode and broadens toward the cathode with further increased bias (lower row of Fig. A.9a). The current-voltage characteristics show no break in the saturation behavior during the transition from cathode- to anode-adjacent domain as indicated by the arrow in Fig. A.9b, in agreement with the prediction from the field-of-direction analysis (Böer and Voss 1968a, b).

As long as the domain is attached to the cathode, the electron density at the cathode and the work function of the cathode are measurable. When, on the other side, the domain is attached to the anode, the hole density at the anode is measurable with the work function of holes from the anode. All this occurs with the same current that has not changed: at the cathode-adjacent domain it is a drift current of electrons, and at the anode-adjacent domain it is again a drift current, but now from the holes:

$$j = j_n = j_p = en\mu_n F_{\text{cathode}} = ep\mu_p F_{\text{anode}}. \quad (\text{A.23})$$

This is very remarkable because the carrier densities and mobilities are different; this holds, however, only as long as high-field domains are present and they adjust in width and field to permit the equality of Eq. A.23.

When plotting the domain width as a function of the bias for the cathode-adjacent (A) and anode-adjacent (B) high-field domains, as shown in Fig. A.10a, one obtains from the slopes and intersections with the abscissa according to Eq. A.22 the fields of the three singular points. These are shown in Fig. A.10b. With the electron mobility

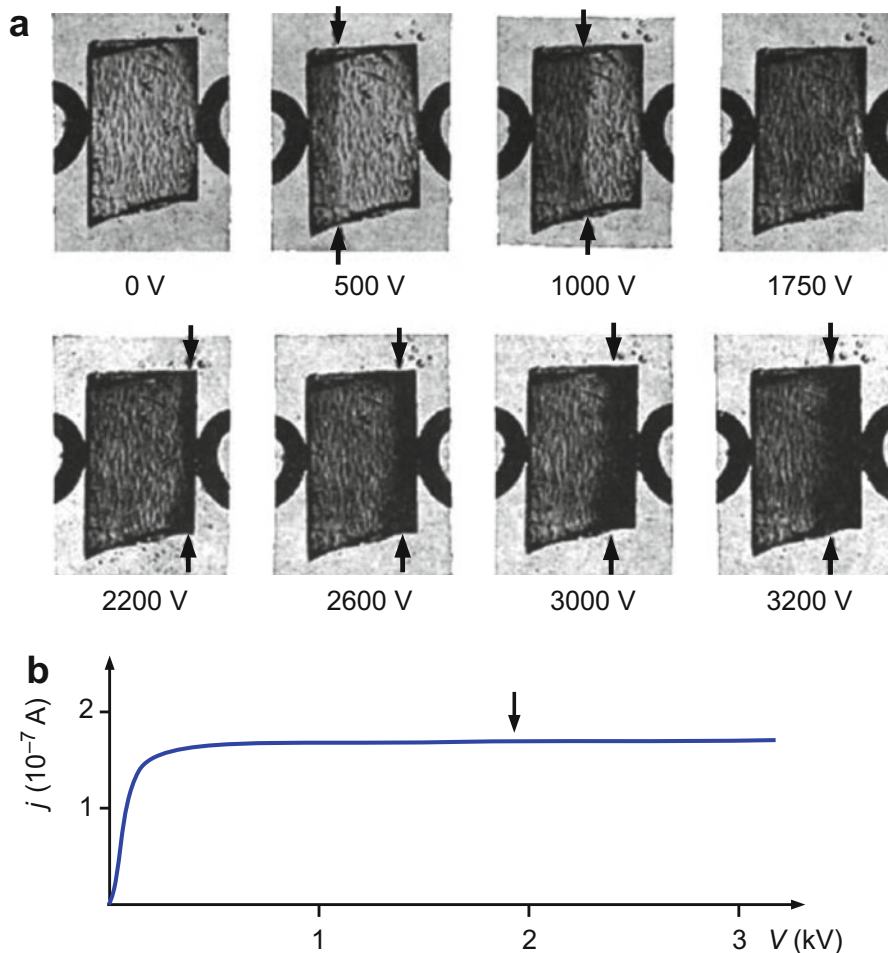


Fig. A.9 (a) Shadow pictures of a CdS platelet (at 210 K) between two electrodes at a distance of 0.36 mm, in monochromatic light (10^{12} photons/cm² s) at the band edge (at 505 nm). The cathode is at the left side, the anode at right; arrows indicate the edge of the high-field domain. The anode-adjacent domain broadens in the opposite direction from the anode toward the cathode indicating that the CdS has changed from *n*-type to *p*-type. (b) Current-voltage characteristics showing current saturation in the entire bias range. The arrow indicates the transition from a cathode- to an anode-adjacent domain and the change of the CdS from *n*-type to *p*-type (After Böer and Voss 1968a)

μ_n and field in the cathode-adjacent domain F_{cathode} or the hole mobility μ_p and field in the anode-adjacent F_{anode} , the carrier densities are then directly obtained from the drift-current curves. Most remarkable is the fact that the drift current at the transition point from cathode- to anode-adjacent domains is also the transition from *n*-type to *p*-type conductivity. With only drift current in the two cases of cathode- and anode-adjacent domains this, hence, requires the equality of Eq. A.23.

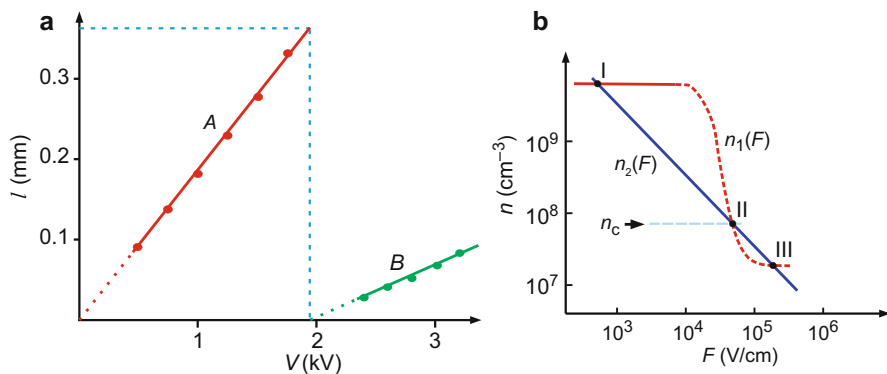


Fig. A.10 (a) Domain width l as function of bias in CdS at 230 K, obtained from the experiment described in Fig. A.9. (b) Singularity points obtained from the slopes of the curves A and B in panel (a) (After Böer and Voss 1968a)

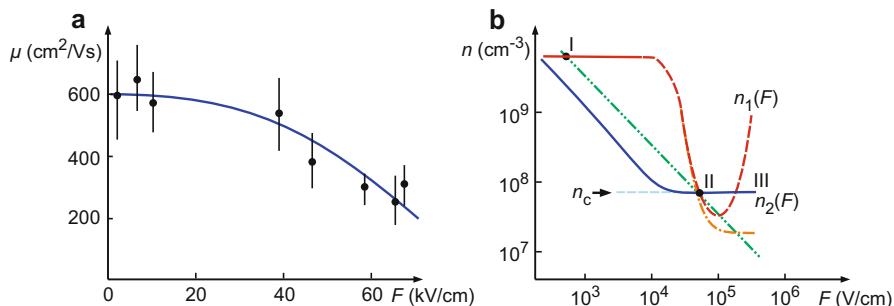


Fig. A.11 (a) Hall mobility in CdS as function of the electric field measured in a high-field domain at 230 K (After Böer and Bogus 1968). (b) Field of directions with corrected $n_2(F)$ and values for the three singularity points obtained by the corrected experimental values. Uncorrected $n_1(F)$ and $n_2(F)$ shown as dash-dotted curve segments

Drift-Velocity Saturation

The field-dependent Hall mobility was measured using high-field domains (Böer and Bogus 1968) with Hall electrodes close to the cathode and applying a sufficient bias so that the cathode-adjacent domain extends well beyond the Hall electrodes. The low-field mobility can be checked by reversing the polarity, thereby placing the Hall probes in the low-field bulk. Different pairs of experimental points were obtained by using different metal electrodes yielding different values of n_c .

The results of these experiments are shown in Fig. A.11. Starting near 20 kV/cm, the mobility decreases with increasing field. Such a decrease follows a $1/F$ behavior at higher fields and causes a saturation of the drift velocity ($\mu F = \text{const}$) as it approaches the rms velocity of the electrons. This results in a bending of the drift-current curve $n_2(F)$ in the field of directions and results in corrected values for the

first and third singular point (Fig. A.11b). The figure also reproduces as dash-dotted curves the results given in Fig. A.10b for comparison.⁷

Workfunction Dependence on Photoconductivity

The field near the metal-semiconductor interface cannot exceed the domain field which is usually < 100 kV/cm, thereby excluding tunneling through the barrier. The carrier density in the domain is therefore a reasonable measure of the metal-semiconductor workfunction according to

$$n_{II} \cong n_c = N_c \exp\left(-\frac{\psi_{MS}^*}{kT}\right), \quad (\text{A.24})$$

where ψ_{MS}^* is the effective workfunction, reduced from the workfunction by the image potential

$$\psi_{MS}^* = \psi_{MS} - \sqrt{\frac{eF_{II}}{4\pi\epsilon\epsilon_0}}, \quad (\text{A.25})$$

with n_{II} and F_{II} the carrier density and electric field, respectively, at the second singular point (Fig. A.7). The image force lowering with fields of $F_{II} \cong 10^5$ V/cm is on the order of 50 meV.

The current over the top of the barrier is carried by Richardson-Dushman emission

$$j = e n_c v_{rms}^* \cong e n_{II} \mu F_{II}, \quad (\text{A.26})$$

with $\mu F_{II} \cong v_{rms}^*$ in the drift-velocity saturation range, this permits the identification of n_c with n_{II} .

When this method is applied to vacuum-cleaved CdS with immediately thereafter vapor-deposited electrodes, one observes with optical excitation rather high saturation currents in reverse bias. This yields workfunctions that are reduced to about one half of their values for vanishing currents which, for a gold electrode or other blocking metals, are on the order of 1 eV.

Figure A.12 shows the linear dependence of domain width as a function of bias, and it indicates a slight increase in domain fields with increasing optical generation rate (from 83 to 122 kV/cm for optical generation rates g_o changing from 4×10^{14} to $5 \times 10^{16} \text{ cm}^{-3} \text{ s}^{-1}$, respectively). Because the carrier density within the domain also increases with increasing optical generation, one thereby observes a trend of a *decreasing workfunction with increasing optical excitation* for all investigated contact metals (Fig. A.12b).

These findings indicate that holes can significantly change the dipole layer at the metal-semiconductor interface. These holes are created by excitation and released by

⁷The shift of the first singular point is not resolved experimentally because of the limited accuracy of the method to determine this point.

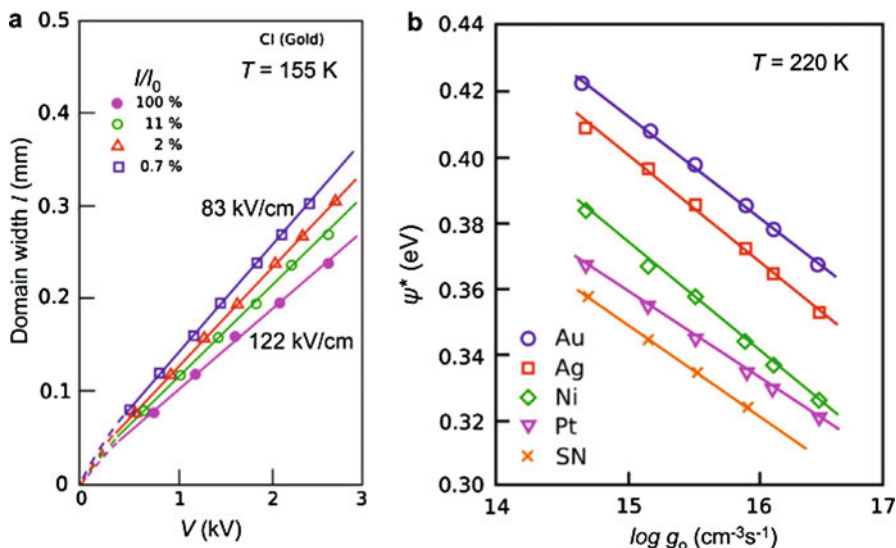


Fig. A.12 (a) Domain width in CdS as a function of the applied voltage with light intensity as family parameter. Slopes yield the domain fields. (b) Effective workfunction as a function of the optical generation rate (After Stim et al. 1971)

field quenching within the high-field domain directly adjacent to the interface and thereby cause the corresponding lowering of the workfunction.

Moving High-Field Domains

In the previous sections, we discussed stationary high-field domains. Under certain boundary conditions the high-field domains become unstable after formation and start to break up and move through the bulk. Moving field domains due to the Gunn effect (Shaw et al. 1979; Shaw et al. 1992) or a recombination instability (Bonch-Bruевич et al. 1975) have been extensively investigated experimentally and theoretically in the 1960s and 1970s. The earliest studies of such *NNDC* and *SNDC* instabilities in semiconductors were published by Böer et al. (1961) for CdS. They discovered moving highfield domains in CdS single crystals (Böer 1959), long before the observation of Gunn domain oscillations in GaAs was reported (Gunn 1963, 1964), and developed the electro-optical method described in Sect. Franz-Keldysh Effect to Directly Observe High-Field Domains above to measure these field inhomogeneities (Böer et al. 1959) based on the first experimental evidence of the Franz-Keldysh effect.

Well-known mechanisms for field-domain formation in bulk semiconductors are provided, for example, by field-quenching in CdS (Böer 1959) and in semi-insulating GaAs (Northrop et al. 1964; Samuilov 1995; Piazza et al. 1997), intervalley transfer (Ridley and Watkins 1961; Gunn 1963; Kroemer 1964),



Fig. A.13 Ring-shaped high-field domain (*arrow*) enclosing a small circular cathode and expanding (a)–(c) in time (pictures taken in 1 s time intervals). After the high-field domain touches the anode at the end of the contact wire at the lower middle left, the ring disappears, a new ring develops at the cathode, and the cycle repeats itself (period length 5 s, contact distance ~ 2 mm).

nonlinear generation-recombination kinetics in the regime of impurity impact-ionization breakdown (Kahn et al. 1991, 1992a, b; Cantalapiedra et al. 1993; Bonilla et al. 1994, 1997; Bergmann et al. 1996) in *p*-Ge, and the photorefractive Gunn effect (Segev et al. 1996; Bonilla et al. 1998).

The Franz-Keldysh effect can most effectively be used to study kinetic effects. Under certain conditions domains can separate from the electrode and move through the device, accompanied by current oscillations, as observed first by Böer et al. (1959) in any solid; see also Böer and Rompe (1960), Böer and Dussel (1967), and Böer and Voss (1968a).

Moving high-field domains that enclose a small circular electrode as a ring indicate the two-dimensional equipotential distribution following the contour of the ring (with equicurrent lines as orthogonal trajectories of the equipotential lines). In the example given on Fig. A.13 (CdS at 250 K) the speed of expansion of the ring is $\sim 10^{-2}$ cm/s. When the ring reaches the anode at one point, it disappears altogether with a sharp increase in current. A new ring then emerges from the cathode with a corresponding current decrease. This motion repeats itself with almost perfect periodicity.

In the stationary high-field domains in *n*-type semiconductors discussed in previous sections the domains either extended from the cathode or, at higher applied voltages with substantially higher fields, emerged from the anode. Under boundary conditions set for the sample shown in Fig. A.13 the high-field domain breaks up and moves toward the anode. Depending on boundary conditions and applied voltage, three kinds of moving high-field domains were observed:

1. Unstable domains that appear near the cathode of an otherwise observed stationary cathode-adjacent domain and move toward the anode while disappearing
 - (a) Before they reach the edge of the previous domain
 - (b) After passing through the edge of this domain
2. Undeformed domains moving from cathode to anode

When moving domains of type 1a or 1b appear, the current oscillates around the value of its stationary saturation, exceeding this value when the moving domain reaches its anode limit, and decreasing below this value after starting near the

cathode. Free-moving domains of type 2 have a much larger field and are maintained by a substantially lower current. For a comprehensive analysis see Böer (2010).

Summary and Emphasis

Whenever the conductivity decreases more than linearly with increasing electric field, a new type of solution of transport and Poisson equations is attained which divides the semiconductor into two parts, one with a high field (the high-field domain) and the other part with a low field. This field distribution can be stationary or nonstationary. In stationarity, the field is essentially constant in each of these regions. Both regions are joined by a narrow step of only a few Debye lengths width.

When the high-field domain moves from one electrode to the other, it disappears when it reaches the electrode, while a new domain evolves from the first electrode and the process repeats itself periodically. Stationary domains are usually observed when the negative differential conductivity is caused by a field-dependent decrease of the carrier density, while no stationary solution can be maintained when the change in conductivity is caused by a mobility that decreases more than linearly with increasing field. Other examples of stationary or moving domains are provided by resonant tunneling in superlattices, real-space transfer in modulation-doped heterostructures, or nonlinear generation-recombination kinetics in *p*-type Ge or semiinsulating GaAs. With increasing bias, stationary high-field domains expand at the expense of the low-field region, until the entire semiconductor is filled by the high-field domain. With further increased bias, a second type of domain with a substantially higher field develops from the opposite electrode and expands as required for absorbing the entire applied voltage.

A relatively simple field-of-direction analysis of the family of solution curves of Poisson and transport equations permits us to obtain a general overview of the different types of domains and to deduce unambiguous information on the field dependence of carrier densities and mobilities, as well as on the carrier density and field at the metal-semiconductor boundary. Stationary high-field domains can be used to measure the workfunction of a metal-semiconductor contact and its dependence on various parameters, providing direct evidence of a reduction of this workfunction with increasing optical generation rates in photoconductors.

The use of a pseudoelectrode by a band of reduced illumination in photoconductors permits the direct measurement of the field-dependent carrier density in semiconductors. Moving high-field domains are slow when related to carrier redistribution over traps and can be very fast when trapping can be neglected in domains caused by a field induced decrease of carrier mobilities. The Gunn effect is an example where such steeper-than-linear decrease of the mobility with increasing field in GaAs is used for high frequency ac generation. High-field domains can be visually observed by the shift of optical absorption edge due to the Franz-Keldysh effect.

High-field domains are a valuable tool to unambiguously investigate the field dependence of important parameters in semiconductors, and as moving domains they are of technical interest for ac generation and the detection of small lattice defects.

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